

THOMAS SESMAT

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EDUCATION

Toulouse Mathematical Institute (non-curricular)

Open-access seminars for researchers

Deep learning, probability, optimization, statistics

Sep 2024 - June 2025

Toulouse, France

National Institute of Applied Sciences

M.S. in Applied Mathematics & Engineering

Relevant coursework: High-dimensional deep learning, AI frameworks, Signal & Image Processing, Machine Learning, Data Analysis, Statistical Modeling, Continuous Optimization, Non-differential Optimization, Advanced Probability, Stochastic Processes

Sep 2021 - Sep 2025

Toulouse, France

Fabert High School

Classe préparatoire aux grandes écoles

Major in Physics & Chemistry

Minor in Mathematics & Informatics

Honorary class

Sep 2018 - Jun 2021

Metz, France

WORK EXPERIENCE

Deezer Research

Research Scientist

June 2026 - Now

Paris, France

ADASP x Deezer Research — Sup. by G. Peeters & G. Meseguer-Brocal

Memorization detection and application to the detection of copyrighted data in audio generation

March 2025 - January 2026

Paris, France

- Critical analysis and improvement of the MTG laboratory's MIRA tools for data replication detection.
- Development of Membership Inference Attack to music on Flow Matching models exploiting memorisation.

ONERA, the French Aerospace Lab — Sup. by J.-L. Farges

Neural Network Internship

Jul - Dec 2023

Toulouse, France

- Theoretical conceptualization and computational development of a mixed-weight neuro-fuzzy network
- Development of binary & continuous optimizer
- Development from scratch without using deep learning/machine learning libraries.

VOLUNTEER EXPERIENCE

Djarama

Civic Service in Agroecology

Feb - Jul 2024

Senegal

- Vegetable cultivation, raising environment and nutritional awareness among children

Junior INSA Service

Member of the Strategic Orientation Committee

Sep 2022 - 2024

Toulouse, France

- Provided guidance on Junior Enterprise strategy, internal audit of treasury department

Junior INSA Service

Treasurer

Sep 2022 - 2024

Toulouse, France

- Accounting and financial management, creation and delivery of treasury training courses

SKILLS AND CERTIFICATIONS

- Programming Languages & tools: Python (PyTorch, TensorFlow, scikit-learn), bash, git
- Technical content writing
- French (Native), English (TOEIC 975/990)

AWARDS & ACHIEVEMENTS

Data Challenge IA Pau

2023

First place (€3000 cash prize)

- Conceptualization and development of a technical solution for a speech and video analysis tool.

ARTICLES

Where Rectified Flow Leaks: Characterising Membership Signals Along the Interpolation Path

2026

43rd International Conference on Machine Learning

- Authors: Thomas Sesmat, Gabriel Meseguer-Brocal, Geoffroy Peeters

SIDE PROJECTS

Pyzyconfig

[In progress]

Python Framework

- Easy-to-use configuration saver for experiments in Python. Useful for small projects that do not require a complex Hydra pipeline.

EasyStart

- Small pipelines that automate commonly used work configurations with a single keystroke (VPN detection required, SSH mounting, opening specific applications, etc.).